

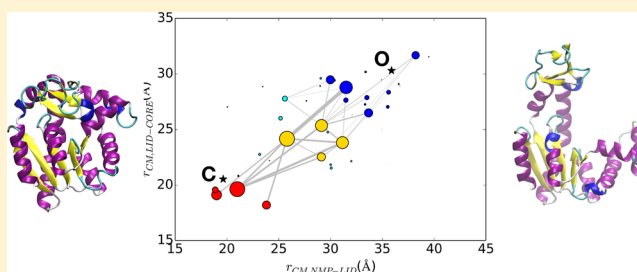
Multiple Pathways and Time Scales for Conformational Transitions in apo-Adenylate Kinase

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Supporting Information

ABSTRACT: The open/close transition in adenylate kinase (AK) is regarded as a representative example for large-scale conformational transition in proteins, yet its mechanism remains unclear despite numerous experimental and computational studies. Using extensive ($\sim 50 \mu\text{s}$) explicit solvent atomistic simulations and Markov state analysis, we shed new lights on the mechanism of this transition in the apo form of AK. The closed basin of apo AK features an open NMP domain while the LID domain closes and rotates toward it. Therefore, although the computed structural properties of the closed ensemble are consistent with previously reported FRET and PRE measurements, our simulations suggest that NMP closure is likely to follow AMP binding, in contrast to the previous interpretation of FRET and PRE data that the apo state was able to sample the fully closed conformation for “ligand selection”. The closed state ensemble is found to be kinetically heterogeneous; multiple pathways and time scales are associated with the open/close transition, providing new clues to the disparate time scales observed in different experiments. Besides interdomain interactions, a novel mutual information analysis identifies specific intradomain interactions that correlate strongly to transition kinetics, supporting observations from previous chimera experiments. While our results underscore the role of internal domain properties in determining the kinetics of open/close transition in apo AK, no evidence is observed for any significant degree of local unfolding during the transition. These observations about AK have general implications to our view of conformational states, transition pathways, and time scales of conformational changes in proteins. The key features and time scales of observed transition pathways are robust and similar from simulations using two popular fixed charge force fields.



INTRODUCTION

Conformational transitions are essential to the function of proteins. These functional motions occur on various temporal and spatial scales in different systems, with the most striking examples being those involved in biomolecular motors and transporters.^{1,2} In more conventional enzymes, large-scale conformational transitions may occur to facilitate substrate binding and product release, and these conformational changes, rather than the chemical step, can be rate-limiting in the catalytic cycle.³ Thus, to fully understand how enzymes and other “biomolecular machines” function, it is crucial to elucidate the mechanism of conformational transitions;^{4–6} meeting this goal will also provide novel design strategies that can be used to manipulate and inhibit the function of specific proteins for bioengineering or medical purposes.

Adenylate kinase (AK) has been a model system for studying conformational transitions in enzymes. AK catalyzes the reversible reaction of AMP phosphorylation by ATP. As shown in Figure 1a, AK has one CORE domain, one ATP-binding domain (LID), and one AMP-binding domain (NMP). Comparison of crystal structures with and without inhibitors^{7,8} suggests that the LID and NMP domains undergo open and close transitions relative to the CORE domain. The opening

transition is found to be rate-limiting in ligand-bound AKs.^{9–11} Fluorescence Resonance Energy Transfer (FRET) and Nuclear Magnetic Resonance (NMR) studies suggested that AK can assume a closed conformation even without the binding of ligand,^{10,11} leading to the proposal that ligand binding may occur in a closed apo state.^{12,13} Numerous experimental^{9–25} and computational^{26–57} studies have been conducted to better define the mechanism of the open/close transition. As summarized in ref 58, however, the mechanism remains unclear, and several distinct proposals have been put forward.

In some studies, the transition was regarded as displacements and rotations of the LID and NMP domains, thus the key step was deemed to implicate the formation of interdomain contacts. Using dynamic importance sampling, Beckstein et al.³⁰ proposed a zipping mechanism that involves the formation of salt bridges between the LID and NMP domains. Coarse-grained (CG) studies using elastic network model^{36,42} and a mixed Go model³³ highlighted regions under strain during domain rotations; it was proposed^{33,36} that the strained region may undergo order–disorder transitions (i.e., “cracking”) to

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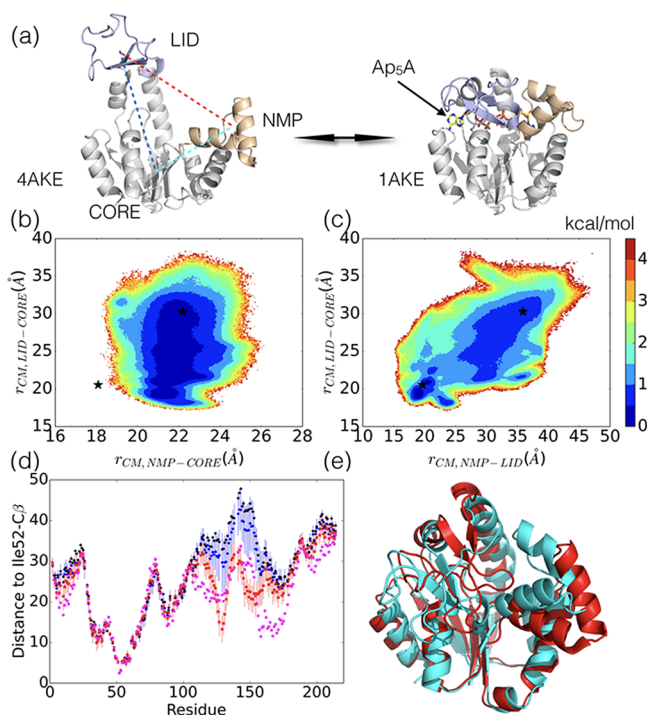


Figure 1. Free energy landscape of apo AK in CHARMM simulations; results for the AMBER force field are in Figure S2. (a) Crystal structures of AK in ligand-free open conformation (PDB ID: 4AKE) and Ap₅A-bound closed conformation (PDB ID: 1AKE). CORE, LID, and NMP domains are colored gray, light blue, and wheat, respectively. Dashed lines show the center of mass (CM) distances between different domains. (b, c) Free energy landscape projected on the 2D space of interdomain CM distances. The stars on the left and right indicate the positions of 1AKE and 4AKE, respectively. (d) Distances of amide proton to C β of Ile52 during the simulations. Black circles show values in open crystal structure (4AKE). Magenta squares show values in closed crystal structure with inhibitor-bound 1AKE. Blue circles show average values in open basin defined in macrostate MSM, and red squares show values in closed basin defined in macrostate MSM. Error bars are ± 1 s.d. Comparison to Figure 3e of ref 10 indicates that the closed ensemble sampled is consistent with the Paramagnetic Resonance Enhancement (PRE) measurements conducted therein; i.e., interpretation of the PRE data does not require the NMP domain to adopt the fully closed conformation. We note that amide protons of all residues except for the C-terminal helix (201–215) are within 30 Å of C β of Ile52 (the labeling site of the PRE experiment). (e) Average closed conformation from the simulation (red) superimposed on 1AKE (cyan).

release the stress and provide entropic stabilization to facilitate the transition. Several recent atomistic simulations and experimental studies of AK variants¹⁶ found hints of local unfolding in the LID-CORE hinge region.^{27,46} In studies using different CG models,^{37,38,40} however, no significant unfolding was observed. The results instead supported the importance of internal stability and dynamics of domains; the latter is consistent with chimera experiments¹⁷ that revealed a broad distribution of residues that influenced functionally important motions in AK.

Since the open/close transition was measured to occur on the μ s–ms time scale,^{11,14} previous computational studies focused on biased atomistic simulations or CG models; free energy simulations^{28,29,43,44,56,59} including string based methods⁴⁸ also rely on a preconceived set of important coordinates and/or initial pathway. In this study, taking advantage of GPU

computing, we conduct multiple rounds of long explicit solvent atomistic simulations to attempt an unbiased analysis of the open/close transition of apo *E. coli* AK. Markov state modeling (MSM)⁶⁰ allows us to identify metastable states, analyze transition pathways, and compute kinetic observables, thus providing novel insights into the mechanism of this prototypical conformational transition.

RESULTS

In the following, we first discuss the structural and dynamic characteristics of apo AK. We then discuss the mechanism of the open/close transition, including major pathways, intermediate states, and time scales. At last, we investigate the origin of multiple time scales and transition pathways. Results from CHARMM and AMBER simulations are generally similar, thus most discussions are based on the CHARMM simulations; results for AMBER are included in the [Supporting Information](#) and briefly summarized in some discussions below.

Closed Conformation of apo AK. The conformational transition is characterized by the opening and closing of LID and NMP domains relative to the CORE domain. Thus, the distances between LID-CORE, NMP-CORE, and NMP-LID domains are good measures of the conformation; they were also closely related to the distances measured from FRET experiments.^{10,11} Based on the conformational ensembles generated from the $\sim 54 \mu$ s explicit solvent simulations, the free energy landscape is projected onto the interdomain center of mass (CM) distances (Figure 1b, c). There is no significant free energy barrier between the open and closed conformations for LID. The NMP domain strongly favors the open conformation, and there is no free energy basin with the NMP domain closed, regardless of the force field used (Figure 1e shows the average closed conformation observed in simulations compared with the ligand bound crystal structure). This is in agreement with the string simulation of Matsunaga et al.,⁴⁶ who also found that the closure of the NMP domain requires ligand binding. Hereafter, the “closed basin” refers to the closed conformation observed in the simulations.

Nevertheless, the distance between NMP and LID domains in the simulated closed basin still reaches that observed in the inhibitor-bound crystal structure (1AKE); this is because the LID domain is significantly rotated toward the NMP domain in the simulated closed structure. In previous studies which concluded that apo AK was able to adopt a closed conformation, the distance monitored was between LID and NMP domains^{10,23,24} or based indirectly on Paramagnetic Relaxation Enhancement effects due to a spin label on the NMP domain¹⁰ (Figure 1d). Therefore, our finding highlights the potential ambiguity in experimental structural characterization when a small number of distances are used. More importantly, our observation suggests that full closure of AK requires the presence of ligands.

Multiple Transition Pathways from MSM Analysis. To study the kinetics and mechanism of the open/close transition, we construct a MSM with 3000 microstates. To facilitate interpretation, the 3000 microstates are coarse grained into 50 macrostates (Figure S10). The open and closed basins are defined as those macrostates near the average open and closed conformations. The major transition pathways between the basins from CHARMM simulations are shown in Figure 2. Four macrostates with large fluxes are identified as the major intermediates; their structures feature a relatively closed LID with a moderate NMP-LID distance. AMBER simulations show

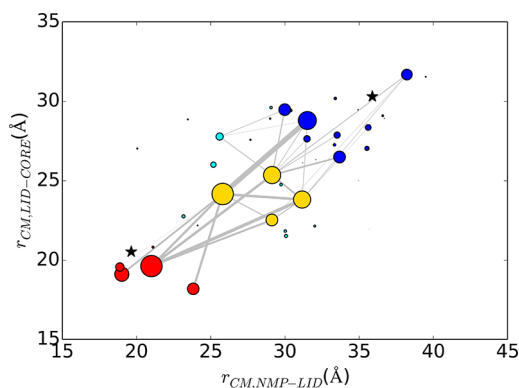


Figure 2. 50-macrostate model of CHARMM simulations and major transition pathways between open and closed basins; results for the AMBER force field are in Figure S11. The size of each dot corresponds to the flux through that state. The width of each line is proportional to the flux. Blue, yellow, and red indicate open, intermediate, and closed macrostates with high fluxes.

similar results except that many pathways pass through one of the major intermediate states (Figure S11b). With both CHARMM and AMBER simulations, no single dominant transition pathway is found. The highest fluxes are 13% and 15% for CHARMM (Figure 3a) and AMBER (Figure S7a), respectively. Therefore, the computational model suggests that the open/close transition involves multiple pathways.

To better understand the structural characteristics of the open, closed, and intermediate macrostates, the corresponding conformational ensembles are analyzed in the contact space

(Figure S13). The characteristic contacts of the open macrostates (O-contacts) are mainly within LID and between CORE, and the adjacent α -helix of the NMP domain (Figure S13a). The characteristic contacts of the closed macrostates (C-contacts) are mainly interdomain contacts; the contacts between LID and the distal helix of the NMP domain are worth noting (Figure S13b). The intermediate macrostates feature contacts between LID and CORE domains and those between NMP and CORE domains, while lacking the contacts between CORE and the distal helix of the NMP domain (Figure S13c). As observed from the superposition of average structures, LID is closed in the intermediate macrostates and further rotated toward the NMP domain in the closed macrostates. The NMP domain in the intermediate and closed macrostates are only slightly more closed compared with the open macrostates (Figure S13d); as emphasized above, the NMP domain in closed macrostates is still open as compared to the inhibitor-bound crystal structure (Figure 1e).

Multiple Transition Time Scales Revealed by MSM Analysis.

In general, multiple pathways do not necessarily imply significantly different time scales. In the case of apo AK, however, we observe distinct transition time scales. Figure 3 shows the distribution of Mean First Passage Times (MFPTs) for microstates in the open and closed macrostates in the CHARMM simulations. The opening transitions occur on time scales of 1.8 and 77.2 μ s. The closing transitions occur on time scales of 12.2, 58.3, and 357.7 μ s. The results for AMBER are shown in Figure S7 and also feature multiple time scales; the opening transitions occur on time scales of 3.0, 10.0, and 17.4 μ s, while the closing transitions occur on time scales of 21.2, 102.2, and 219.5 μ s. The overall MFPT between the open/

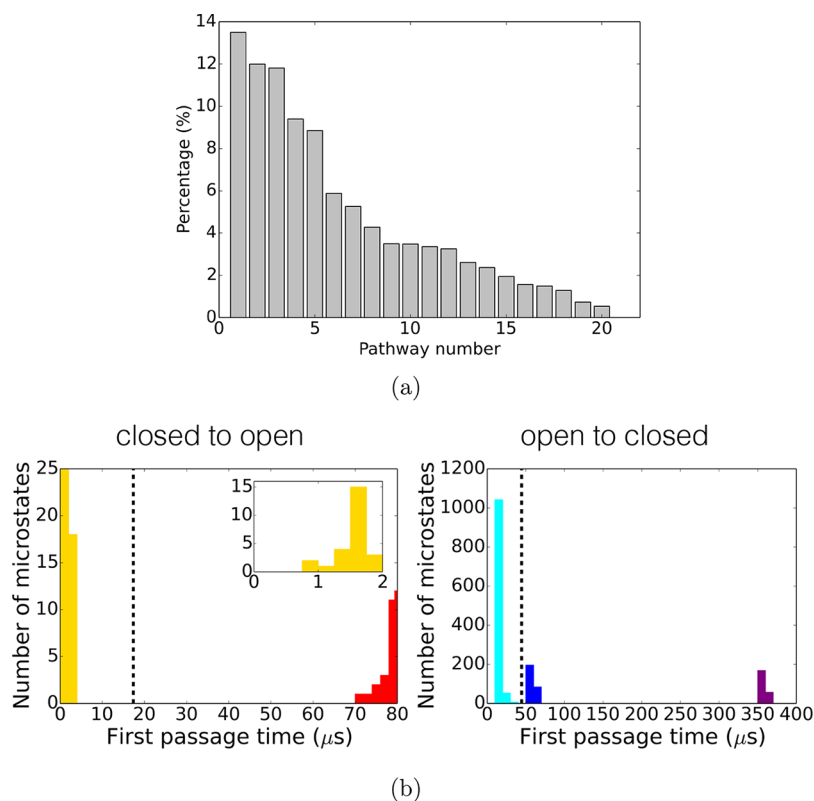


Figure 3. Multiple pathways sampled in the CHARMM simulations feature multiple time scales; results for AMBER simulations are in Figure S7. (a) Percentage of total flux in each of top 20 transition pathways. (b) Mean First Passage Time (MFPT) distribution of microstates in the closed and open basins. The vertical dashed lines indicate the average MFPT.

close basins for CHARMM is $17.3\ \mu\text{s}$ for the opening transition and $44.5\ \mu\text{s}$ for the closing transition; the corresponding values for AMBER are 9.0 and $36.4\ \mu\text{s}$, respectively. The estimated transition time scales do not change significantly when different numbers of macrostates are assigned to the open/close basins in the MFPT calculations (Table S1), and the MFPTs of microstates within the same macrostate are similar (Figure S8), further supporting the robustness of the kinetic coarse-graining procedure.

The multiple time scales for the open and close transitions are also evident in unbiased MD trajectories, which show transition events occurring on time scales ranging from nanoseconds to microseconds (Figure S9). Using the 3000 state MSM, milliseconds-long pseudotrajectories are generated using a kinetic Monte Carlo scheme. Movie 1 (link is in the Supporting Information) shows a 1.5 ms long trajectory with conformational transitions on the time scales of tens of microseconds to several hundreds microseconds. Movie 2 (link is in the Supporting Information) is $9\ \mu\text{s}$ long with a higher time resolution, showing transitions from hundreds of nanoseconds to a couple of microseconds. From these milliseconds pseudotrajectories, open/close transition time scales are also estimated from the lifetime distribution of open/closed states. The analysis reveals multiple time scales that range from tens of nanoseconds to microseconds (see detailed discussion of lifetime analysis in the Supporting Information).

Origin of Multiple Transition Time Scales and Pathways. To investigate the origin of the multiple pathways and time scales, we compare the fast and slow pathways in Figure S12; in both CHARMM and AMBER simulations, the slow pathways have relatively small flux and are not among the pathways with top 20 fluxes. In terms of macrostates, the slow pathways may go through the same intermediates as the high flux fast pathways, but in general the pathways appear distinct for different time scales. For example, with the CHARMM simulations, the slow opening transition uniquely involves macrostates 1 and 4, while the slow closing transition uniquely involves macrostates 2 and 18; the equilibrium populations of these macrostates are not small. It would be of interest to conduct further analysis, such as finite temperature string simulations, to compare the free energy profiles along these distinct pathways and identify the factors that dictate the different time scales; considering the complexity of such calculations, we leave such analyses to the future.

To probe the connection between structural and kinetic properties of different conformations, we calculate the committor values for each microstate transitioning to the open and closed basins. Ten microstates near the average open/closed conformations are used to define the open/close basins, and the committor values show the probability for each microstate to reach the closed basin before reaching the open basin (Figure 4). The committor values for the microstates near the open basin are generally very small, as expected. By contrast, microstates near the closed basin have a broad range of committor values, instead of having values close to one as a simple model would imply. This observation suggests that microstates near the closed basin have a high degree of kinetic heterogeneity. These trends are not sensitive to the number and selection of microstates used to define the open/close basins or to the force field used; the only notable difference is that AMBER simulations lead to a separate cluster of microstates with committor values of around 0.4 (Figure S16), while a cluster with values around 0.3 is observed with

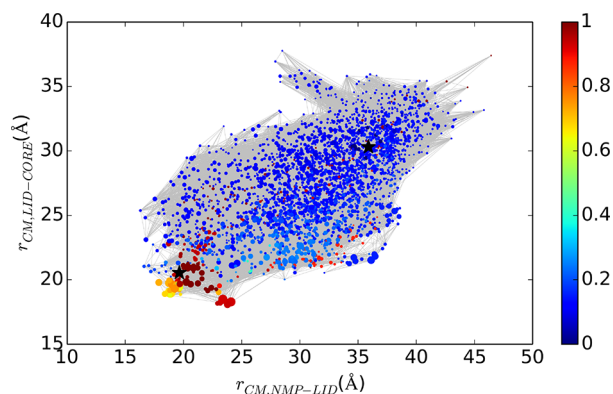


Figure 4. Committor probability values of each microstate toward the closed basin in the 3000-state MSM from CHARMM simulations; results for AMBER simulations are in Figure S16. Microstates are projected onto the NMP-LID and LID-CORE interdomain CM distances. The size of each dot is proportional to the population of the corresponding microstate. The black stars indicate the locations of the open/closed crystal structures.

CHARMM, and it is less separated from microstates with lower committor values (Figure 4). The microstates in these clusters with intermediate committor values generally have a small LID-CORE distance and large NMP-LID distance.

The average structure of states with similar MFPTs is not significantly different from that of states with different MFPTs (Figure S17). As shown in Figure 5, structures of microstates near the closed basin may have similar fractions of C-contacts (and RMSD values relative to the averaged closed conformation) but substantially different committor values. The

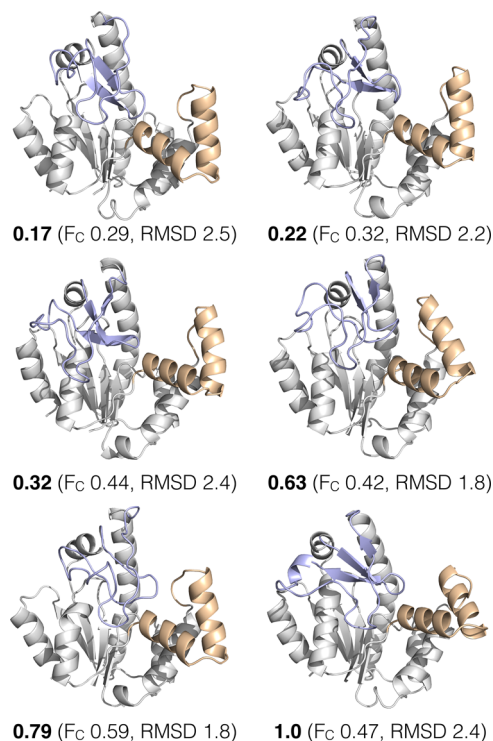


Figure 5. Representative structures in closed microstates with various committor values toward the closed basin. Bold numbers are committor values. F_c is the fraction of C-contacts. RMSD (in Å) is relative to the average closed conformation.

fraction of C-contacts and O-contacts is not a good predictor of the transition rate either. In CHARMM simulations (Figure S18), for instance, the closed microstates with a faster transition to the open states generally have a *higher* fraction of C-contacts compared with the microstates with slower transitions. This unexpected trend indicates the contacts are not equally important in determining the transition kinetics. To identify the interactions that are strongly correlated to transition time scale, the mutual information between the formation of different interactions and transition time scales is analyzed. Figure 6 shows the hydrogen bonds, salt bridges, and

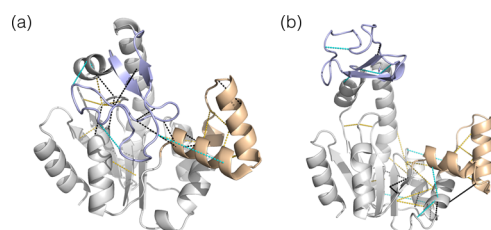


Figure 6. Interactions of highest mutual information with transition time scales in CHARMM simulations; results for AMBER simulations are in Figure S19. For a detailed list of residues, see Tables S2–S5 in the Supporting Information. (a) Top 30 interactions of highest mutual information with closed to open transition time scales mapped onto the average closed structure. (b) Top 30 interactions of highest mutual information with open to closed transition time scales mapped onto the open crystal structure (4AKE). Black: hydrogen bond; cyan: salt bridges; yellow: hydrophobic contacts. When two residues are involved in more than one interaction of high mutual information, the color is based on the interaction of the highest mutual information.

hydrophobic contacts of highest mutual information with transition time scales (Tables S2–S5, Supporting Information) in CHARMM simulations. Out of the 17 residue pair interactions that influence both opening and closing time scales, four are intra-LID interactions, three are intra-NMP interactions, three are LID-CORE interactions, six in the LID-CORE hinge region, and one in the NMP-CORE hinge region. Out of these 17, seven residues pairs are from CHARMM simulations, and 10 are from AMBER. Since CHARMM and AMBER simulations lead to distinct transition pathways (compare Figure 2 and Figure S11b, Supporting Information), it is not surprising that distinct pairs of contacts are identified from the MI analysis. Nevertheless, CHARMM and AMBER results consistently show the importance of intradomain contacts and interdomain hinges to the transition kinetics (though we caution that the partition of protein into domains is not without uncertainty, especially those residues at the domain boundaries); specific pairs such as residues 123–159 and 156–159 in fact emerge consistently in both CHARMM and AMBER simulations as intradomain contacts. Therefore, while we would be cautious about the precise predictions that can be made from our simulation studies considering the scale of sampling and use of empirical force fields, we highlight these residues in the LID domain as possible targets for further mutation studies that probe residues critical to the open/close transition kinetics.

Limited Degree of Secondary Structure Disruption during Transition. It was proposed^{16,36} that local unfolding (“cracking”) of the hinge regions might help to release local strain and provide entropic stabilization for the large-scale transition. All macrostates in the 50-state MSMs are scrutinized

for secondary structure stability. As shown in Figure 7, the major intermediate macrostates are largely intact in terms of

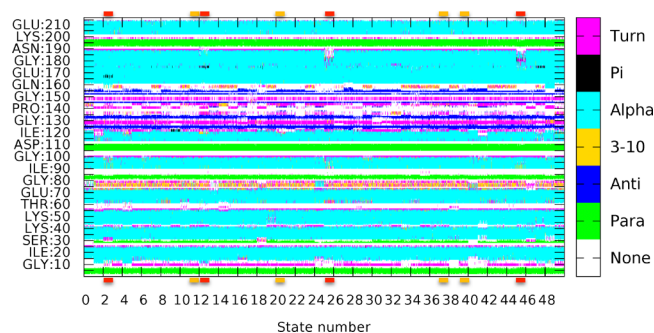


Figure 7. Secondary structure analysis of the 50 macrostates from CHARMM simulations; results for AMBER simulations are in Figure S20. The yellow bars indicate the macrostates identified as major intermediate states (Figure 2), and the red bars indicate the macrostates that have signs of secondary structure disruption. For representative structures with secondary structure distortions, see Figure 8.

secondary structures. In some states, the middle region of the long hinge α -helix becomes a π -helix, and the C-terminal of the helix shows local unfolding. The representative structures in these states (Figure 8), however, do not resemble the proposed “cracking” in position (Figure S21) or amplitude.³³

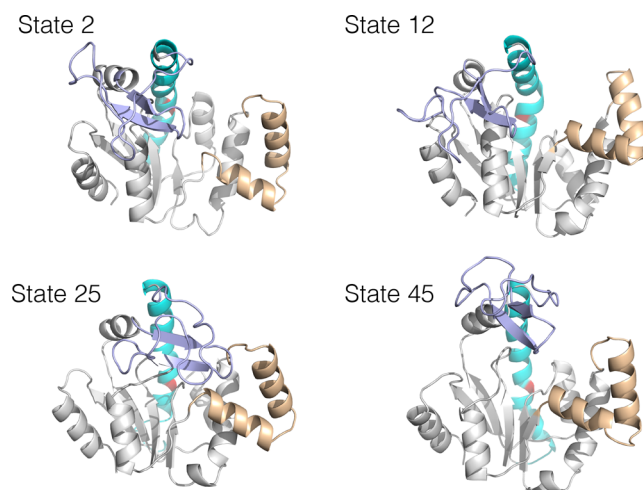


Figure 8. Representative structures for the macrostates that have secondary structure disruption in CHARMM simulations (macrostates labeled with red bars in Figure 7). The CORE, LID, and NMP domains are colored gray, light blue, and wheat, respectively. The helix hinge between CORE and LID domains is colored cyan with Pro177 colored red.

DISCUSSION AND CONCLUDING REMARKS

There is continuing debate about whether AK binds substrate with an “induced fit” mechanism or by “conformational selection”. As discussed in ref 4, these terminologies were originally introduced by Koshland et al.⁶¹ and Monod et al.⁶² in different thermodynamic models to describe allosteric coupling in protein complexes. In recent literature, however, they were often used to describe the kinetic pathways of ligand binding in proteins.⁶³ In induced fit, ligand binding occurs in the open conformation of the protein and precedes closure. In a

population shift or conformational selection model, closure of the active site occurs in the absence of the ligand, and ligand binding occurs in the closed conformation. The pathway that dominates may depend on ligand concentration and protein dynamics.^{63,64} Among previous studies of AK, some supported the induced fit mechanism,^{25,33,46} while others endorsed the conformational selection mechanism^{12,26,65} due mainly to the different characterization of the conformational ensemble of apo AK. A recent experimental study¹³ trapped AK in the closed catalytically active state using a disulfide bond and demonstrated the feasibility of a conformational selection mechanism, although it was argued that the wild-type AK likely uses both induced fit and conformational selection.

Without mixing the different usage of induced fit and conformational selection (Figure S30, Supporting Information), we note that in this study the closed state in apo AK is observed to differ from that in the ligand bound form, and although LID samples closed configurations, the NMP domain remains largely open in the absence of stabilizing interactions mediated by the substrate. Therefore, while only apo simulations are conducted here, our results suggest that full closure of AK likely follows ligand binding to the NMP domain, while ATP binding may occur with either an open or closed LID domain.¹³ This conclusion is in contrast with the previous interpretation of experimental data that the apo state samples fully closed conformation,^{10–13} highlighting the limitation of low-dimensional structural characterization with measurements such as FRET,^{10,11} PRE,¹⁰ and cross-linking.¹³

The MSM analysis reveals that the open/close transition occurs on multiple time scales ranging from several microseconds to hundreds of microseconds through multiple pathways. The open basin is kinetically homogeneous, while the closed basin is observed to be kinetically heterogeneous. The latter indicates the complexity of the local interactions that stabilize the closed conformations. The transition from one closed metastable state to another requires breaking a subset of interactions and forming new ones. Since breaking a subset of interactions can lead to an opening of the conformation while forming the contacts specific to another closed metastable state is stochastic, some closed metastable states are kinetically closer to the open states rather than other members of the closed basin.

It is not always straightforward to compare the computed time scales to experimental data due to significant discrepancy in the latter. We focused on *E. coli* apo AK in the absence of either nucleotides or Mg^{2+} . An NMR study of Shapiro et al.⁶⁶ found a LID and NMP domain opening rate of $1.9 \times 10^7 \text{ s}^{-1}$, which corresponds to a time scale of 52 ns. On the other hand, single molecule FRET experiments observed much slower time scales for LID with an open and closed time scale of 8.3 and 4.5 ms, respectively.¹¹ An NMR relaxation dispersion and FRET study of AK from the thermophile *Aquifex aeolicus* in the apo form measured the motions of LID and NMP simultaneously and reported an open and closed time scale of 150 and 500 μs , respectively, at room temperature;¹⁰ the time scales for the mesophilic *E. coli* AK are expected to be shorter.¹⁷ Apparently, the measured time scales from different experiments span rather broad ranges, which is due in part to the different labeling protocols and low-dimensional nature of the structural observable(s).^{30,48}

In this study, MSM analysis shows that the opening transitions occur on time scales of several microseconds to tens of microseconds; the closing transitions occur on time

scales of tens of microseconds to hundreds of microseconds; these computed time scales suggest that the open state is slightly lower in free energy than the closed state, in agreement with experimental observations.^{10,12} By fitting the lifetimes (or dwell/waiting times) of open and close states using pseudotrajectories constructed using a kinetic Monte Carlo protocol, multiple time scales ranging from tens of nanoseconds to several microseconds are observed (Figures S23–29). The lifetime analysis uses solely interdomain distances to define transitions; thus, it may have led to faster time scales compared to MSM analysis, which considers many structural features to define macrostates and transitions among them. The fastest time scale identified from our analysis is close to that measured by NMR spin relaxation,⁶⁶ while the slowest ones are on the similar $10^2 \mu\text{s}$ time scale inferred from the *Aquifex aeolicus* studies.¹⁰ It is possible that only the slow transitions were captured in FRET experiments due to the limited time resolution; indeed, the longer first passage times from the simulations (Figure 3) appear to be in closer agreement with experimentally measured values^{10,12} than the overall MFPTs. Another factor worth noting, in addition to the different flexibilities (thus transition rates) of thermophile and mesophile AKs,¹⁷ is that the simulations here use the TIP3P water model, which is known to underestimate viscosity by a factor of ~ 2 ;⁶⁷ since solvent motion is expected to be involved in domain transitions,⁶⁷ it is possible that the long time scale is underestimated in our study by a factor of 2–3.⁶⁸ Further analysis is required to understand whether the transition rate is dominated by solvent friction or the internal friction of the protein.⁵

One general mechanistic question regarding large-scale conformational transitions is what factors dictate the transition rate/time scale. A more traditional view of large-scale transitions is that they occur largely through displacement/rotation of semirigid domains mediated by hinge residues;^{4,14} this is indirectly supported by many normal-mode analysis studies.^{69–71} In this work, contacts associated with a set of hinge residues are observed to correlate with the transition kinetics, further supporting the notion that properties of these residues facilitate slower domain motions.¹⁴ An alternative view, by contrast, emphasizes the role of reversible order–disorder transitions (i.e., local unfolding or “cracking”), which ties protein (local) stability and dynamics in an elegant framework.^{16,20,21,33,34,36,72,73} In the specific case of apo AK, our simulations do not point to any compelling evidence for significant local unfolding (Figure 7). Although it is possible that sampling the pathways that involve a significant degree of local unfolding requires even more extensive simulations than carried out here (e.g., it is possible that the empirical force fields used here might be “native-centric”, i.e., overstabilizes compact conformations⁷⁴), the fact that the observed time scales in our work are comparable or faster than experimental values suggests that significant unfolding is not required for rapid conformational transitions in apo AK. Nevertheless, the mutual information analysis of interactions and transition time scales identifies the importance of both interdomain and intradomain interactions. For instance, a set of intra-LID domain interactions appear to influence both opening and closing transition time scales. This observation is qualitatively consistent with a previous study¹⁷ using chimeras of thermophilic and mesophilic *Bacillus* AKs, which found that the internal properties of domains were correlated with the enzyme kinetics, which presumably reflected the rate of the

opening transition. Therefore, the importance of internal domain properties to conformational transition kinetics may reflect the relevance of specific intradomain interactions rather than the necessity of domain unfolding during the transition.

Finally, we note that CHARMM and AMBER simulations largely result in similar observations, such as the conformational features of key macrostates, the kinetic heterogeneity of the closed basin and multiple transition pathways and time scales. In the future, it is worthwhile studying the open/close transition in the presence of the ligands and Mg^{2+} , although the strong coupling of ligand binding and conformational transition poses important technical challenges that need to be overcome.⁷⁵

■ COMPUTATIONAL METHODS

Simulation Protocol. Due to the long (μs – ms) time scale nature of the open/close transition of AK, we use a multiround strategy that integrates implicit and explicit solvent simulations. First, implicit solvent simulation is used to explore the conformational space and generate “seed” conformations for subsequent explicit solvent simulations. Two trajectories are initiated using the AMBER ff99SBnmr1 force field⁷⁶ and the GB7 implicit solvent model,⁷⁷ one starting from the crystal structure of ligand-free form in the open conformation (PDB ID: 4AKE⁸) and the other starting from the crystal structure of inhibitor-bound form in the closed conformation (PDB ID: 1AKE⁷). Open/close transitions are observed in both simulations ($\sim 2.5 \mu\text{s}$ each), and 28 “seed” conformations are selected such that they cover the space with AK in different intermediate conformations. The 28 “seed” conformations together with the two crystal structures are subject to the first round of explicit solvent simulations. Two independent sets of simulations using the CHARMM36⁷⁸ and AMBER ff99SBnmr1⁷⁶ force fields are carried out with explicit solvent for the purpose of comparison and cross-validation. About 30 μs trajectories are generated for each force field. After analyzing the ensemble and building a 50-state MSM, it is observed that the region near the closed conformation is not well sampled. Thus, the second round of sampling is carried out as follows. Five trajectories are initiated from 1AKE. Ten trajectories are initiated using snapshots in the less sampled region near the closed conformation. Ten trajectories are initiated using randomly selected structures from the 10 least populated states in the MSM. Accordingly, an additional 15 μs trajectories are generated for each force field. After analyzing the ensemble from the first two rounds, a third round of sampling with 16 trajectories is carried out to sample the poorly covered regions thus far. After three rounds of sampling, a total of about 54 μs trajectories are generated for each force field, with single continuous trajectories ranging from 350 ns to $\sim 1.2 \mu\text{s}$. AMBER14 MD package with graphics processing units⁷⁹ is used in all simulations. The convergence of the simulations are indicated by the convergence of free energy landscape and implied time scales in MSMs (Figure S1).

In the implicit solvent simulations, the initial structures are minimized and equilibrated using the GB7 generalized Born model. No cutoff is applied to the nonbonded interactions. The salt concentration of 0.15 M NaCl is imposed with the Debye–Hückel limiting law for ionic screening of electrostatic interactions.⁸⁰ All bonds involving hydrogen atoms are constrained using the SHAKE algorithm.⁸¹ An integration time step of 2 fs is used. The temperature is maintained at 310

K using Langevin dynamics with a collision frequency of 20 ps^{-1} .

In the explicit solvent simulations, CHARMM_GUI⁸² and CHAMBER⁸³ are used to generate input files for simulations using the CHARMM36 force field. TIP3P water model⁸⁴ is used with periodic boundary conditions. The system size is about $79 \text{ \AA} \times 79 \text{ \AA} \times 79 \text{ \AA}$, maintaining the closest distance between protein atoms and water box edges to be greater than 10 $\text{\AA}$. Here, 0.15 M NaCl is added to the system after neutralizing with counterions. The nonbonded interaction cutoff is set to be 12 $\text{\AA}$; the missing long-range van der Waals interaction is approximated with a long-range continuum correction.^{85,86} Electrostatic interactions are calculated with particle mesh Ewald⁸⁷ with a grid spacing of about 1.0 $\text{\AA}$. Energy minimization is done with 3000 steps of steepest descent followed by 97,000 steps of conjugate gradient. The protein is fixed initially and then relaxed to minimize the energy of the entire system. Langevin dynamics is used to heat the system to 310 K gradually in 20,000 steps with a collision frequency of 1 ps^{-1} . Here, 1 ns of NPT simulation is then carried out to equilibrate the density of the system. Production simulations are carried out in NVT ensemble at 310 K using Langevin dynamics with a collision frequency of 1 ps^{-1} . All bonds involving hydrogen atoms are constrained using the SHAKE algorithm, allowing an integration time step of 2 fs. Snapshots are saved every 20 ps, resulting in about 2.7 million conformations for each force field. The explicit solvent AMBER simulations are generated using a similar procedure.

Building Markov State Models. MSMBuilder2⁸⁸ is used to build the Markov state models. A different number of microstates are generated using the hybrid k-centers/k-medoids clustering method⁸⁸ based on the root mean squared deviation (RMSD) of backbone atoms ($\text{C}\alpha$, C, N, O) and $\text{C}\beta$. The transition probability matrix can be obtained by normalizing the symmetrized count matrix. The implied time scales for a specific lag time are calculated as

$$\kappa(\tau) = \frac{-\tau}{\ln[\mu(\tau)]} \quad (1)$$

where κ is a relaxation time scale, τ is the lag time, and $\mu(\tau)$ is an eigenvalue for the transition matrix $\mathbf{T}(\tau)$. The implied time scales plateau at longer lag times. After calculating implied time scales for 1000-, 2000-, 3000-, 4000-, 6000-, and 8000-state microstate MSMs, the slowest time scales converge when using 3000 or more states. Thus, the 3000-state models are chosen for further analysis. Since the implied time scales plateau at around 12 ns, 12 ns is chosen as the lag time to build the final models (Figure S3). The 3000-state MSMs are verified using the Chapman–Kolmogorov test by comparing time evolution of the probability of remaining in a certain state calculated from the model and that from the MD trajectories (Figure S4).

To interpret the MSMs, the 3000-state microstate MSMs are coarse grained to 50 macrostates using the Bayesian agglomerative clustering engine (BACE),⁸⁹ which has been shown to be a robust method for coarse graining MSMs with large number of microstates.⁹⁰

Markov State Model Analysis. Transition path theory is used to compute the transition pathways between the open and closed basins. Commitor probability values for the microstates in the 3000-state MSMs are calculated using methods described in the literature.^{91,92} Since the coarse grained models tend to underestimate the time scales, the MFPTs between the open

and closed macrostates are calculated based on the 3000-microstate MSMs as previously described.^{93,94}

Structural Analysis. Hydrogen bonds are defined by a distance cutoff of 3 Å between donor/acceptor heavy atoms and an angle cutoff of 135 degrees. Salt bridges are defined by a distance cutoff of 4 Å between the center of mass of the oxygens in the acidic side chain and that of the nitrogens in the basic side chain. Hydrophobic contacts are defined by a distance cutoff of 5 Å between the side chain center of mass of two hydrophobic residues. Native contacts are defined as contacts where the distance between C β atoms of two residues (C α for Gly) is within 8.0 Å, excluding adjacent residue contacts ($|i - j| < 5$). Characteristic contacts are those that have a probability difference greater than 0.2 between the open and closed macrostates (Figure S15, Supporting Information). Cpptraj⁹⁵ is used to process the trajectories, and AMBER Tools⁹⁶ are used to calculate the distances. Secondary structures are determined using the DSSP method⁹⁷ based on backbone amide and carbonyl atom coordinates.

Mutual information (MI)⁹⁸ of interactions is defined as $MI(Y) = -\sum_{y \in Y} P(y) \ln P(y) - \sum_{x \in X} P(X=x) (-\sum_{y \in Y} P(Y=y | X=x) \ln P(Y=y | X=x))$. $Y = 0, 1$ denotes the existence of an interaction; $X = 0, 1$ indicates fast and slow time scales, respectively, when the MFPT distribution is bimodal; $X = 0, 1, 2$ indicates fast, medium, and slow time scales, respectively, when MFPT distribution is trimodal. The contacts that feature high MI values are summarized in Tables S2–S5.

Generation and Analysis of Pseudotrajectories from MSM. A kinetic Monte Carlo scheme is used to generate pseudotrajectories based on the transition probability matrix of the 3000-microstate MSMs. After randomly selecting a starting state, the next state is chosen according to the transition probability matrix, and the process is propagated. A random snapshot is chosen from the conformations that are assigned to each corresponding state. A detailed description of the procedure can be found in literature.⁹⁹ For each force field, 10 independent trajectories are generated by randomly selecting 10 different starting states, each 120 ms long. Reaching equilibrium is detected by comparing the population of each state obtained in the trajectories with the equilibrium population calculated from MSM (Figure S22, Supporting Information).

Distances of interest are then calculated for the generated trajectories, such as the distance between CM of the LID and NMP domains. The distribution of the distances are approximated by the summation of two Gaussian densities. The values where the density difference between these two Gaussians is equal to 0.015 are used to separate the open state, closed state, and a transition zone. This procedure mimics the data processing in a FRET study.¹⁰ Varying ± 0.005 in the choice of the density difference does not have significant influence on the calculated time scales. From the trajectories, lifetime (or dwell/waiting time) distributions¹⁰⁰ of the closed and open states can be obtained. The lifetime distributions are fitted with a sum of exponential decays, and the corresponding time constants are obtained.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.7b01064.

Analysis for the convergence of simulation, validation of Markov state models, and additional analysis of the transition mechanism with committor distribution, contact distributions, mutual information analysis, and lifetime analysis using a kinetic Monte Carlo protocol, and also links to movies for pseudotrajectories generated using kinetic Monte Carlo at different time scales. (PDF)

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Notes

The authors declare no competing financial interest.

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